

DNS Pharming — Full Case Study Reference

A categorised reference of real-world DNS pharming, spoofing, cache poisoning, and hijacking incidents, grouped to align with the lab's three attack techniques, plus two broader categories for infrastructure-level and nation-state incidents.

Category 1 → Lab Task 1 HOSTS file	Category 2 → Lab Task 2 LAN spoofing	Category 3 → Lab Task 3 Cache poisoning	Category 4 Infrastructure & BGP hijacks	Category 5 Nation-state & registrar hijacks
---	---	--	--	--

1 Local / Host-Level Manipulation

Maps to Task 1 — modifying the HOSTS file / local resolver settings

DNSChanger Malware	2007-2011
Malware silently rewrote infected machines' DNS server settings — the same effect as editing <code>/etc/network/interfaces</code> in your lab, just delivered remotely and covertly. Millions of machines worldwide were redirected to attacker-controlled resolvers for ad fraud and further phishing, until the FBI's "Operation Ghost Click" took the infrastructure down.	

2 LAN-Level Spoofing & Router Pharming

Maps to Task 2 — sniffing a query and racing the real DNS server with a forged reply

Brazilian Home-Router Pharming	2011	Brazilian Bank DNS Hijack	2018
Attackers exploited default router passwords and poisoned local ISP-level DNS caches to redirect online banking customers to convincing phishing clones. Thousands of victims entered banking credentials, often over weeks without noticing.		A major Brazilian bank had its entire DNS hijacked for over five hours, silently redirecting essentially all customer traffic to attacker-controlled infrastructure. One of the most complete real "own the resolver" pharming events on record, though under-reported.	

3 Server / Resolver-Side Cache Poisoning

Maps to Task 3 — poisoning the DNS server so every subsequent query is affected

The Kaminsky Bug	2008	ISP Cache Poisoning → Malware	Nov 2011
Dan Kaminsky discovered that predictable DNS transaction IDs and source ports made blind cache poisoning far easier than assumed — poison the resolver once, and every subsequent user of that server gets the spoofed answer. Triggered a coordinated, multi-vendor emergency patch across nearly every DNS implementation.		Attackers poisoned an entire ISP's DNS cache, redirecting users through a malware-installation step before sending them on to the site they had actually requested — making the attack far less noticeable.	
SAD DNS (CVE-2020-25705)		2020	
UC Riverside researchers showed that ICMP rate limiting in Linux kernels could be exploited as a side channel to infer which source port a resolver was using, reducing entropy enough to make cache poisoning practical again — even after post-Kaminsky port-randomisation defences. Affected an estimated 34% of open resolvers tested.			

4 BGP + DNS Infrastructure Hijacks

Attacking the routing/DNS infrastructure itself, usually for direct financial theft

MyEtherWallet DNS Hijack	Apr 2018	BlackWallet DNS Hijack	Jan 2018
Attackers hijacked BGP routes for Amazon's Route 53 DNS service, then served fake DNS answers for myetherwallet.com, redirecting users to a phishing clone in Russia. ~215 ETH (~US\$152,000) stolen within ~2 hours before control was regained. See detailed timeline sheet.		DNS hijack via a compromised Amazon Route 53 configuration redirected BlackWallet.com users to a phishing clone. Over US\$400,000 in Stellar Lumens (XLM) was stolen.	
EtherDelta DNS Hijack		Dec 2017	
A similar DNS hijack redirected users of the EtherDelta cryptocurrency exchange to a phishing clone shortly before Christmas 2017. Total losses were never fully confirmed publicly.			

5 Nation-State & Registrar-Level Campaigns

Pharming used for espionage or censorship rather than direct financial theft

Sea Turtle Campaign	2017-2019	DNSpionage	2018-2019
Cisco Talos documented a state-sponsored operation that compromised DNS registrars and ccTLDs directly to hijack DNS for 40+ organisations across 13 countries, enabling large-scale credential harvesting against government and infrastructure targets.		A related Middle East-focused campaign that compromised DNS records of government agencies and private companies, redirecting traffic through attacker infrastructure to intercept credentials.	
China's "Great Firewall" DNS Leak			2010
China's internal DNS censorship infrastructure leaked outward onto the wider internet, causing international users querying Facebook, Twitter, and YouTube to be redirected to Chinese censorship servers — a rare case of a national poisoning system spilling over accidentally.			

TEACHING NOTE

Some frequently cited "DNS hijack" headlines (e.g. brief outages where a major site briefly resolved incorrectly) were actually **registrar account compromises** rather than classic resolver-side cache poisoning or LAN spoofing — still DNS-record tampering, but a distinct technique from Tasks 2 and 3. Worth flagging this distinction explicitly, since students often conflate "DNS records changed at the registrar" with "a resolver's cache was poisoned."